Predicting Psychiatric Illness Risk from Neuroimaging–Genetics Associations

Master Thesis – Progress Update

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Data

Raw Participant-Level Data

Two matrices share matched participant rows ($N_1 \dots N_n$):

Structural MRI ($n \times m$) **Genotype** ($n \times l$)

$$\mathbf{X}_{\text{MRI}} = \begin{pmatrix} \text{MRI}_{1,1} & \cdots & \text{MRI}_{1,m} \\ \vdots & \ddots & \vdots \\ \text{MRI}_{n,1} & \cdots & \text{MRI}_{n,m} \end{pmatrix} \qquad \mathbf{X}_G = \begin{pmatrix} G_{1,1} & \cdots & G_{1,l} \\ \vdots & \ddots & \vdots \\ G_{n,1} & \cdots & G_{n,l} \end{pmatrix}$$

- n = number of participants (European ancestry cohort)
- m = MRI phenotypes (volume, area, thickness, intensity, contrast per brain region)
- l = genotyped SNP variants

GWAS Transformation

For every SNP i and MRI phenotype j, a GWAS collapses the n participants into a single z-score:

$$\text{GWAS}(\mathbf{X}_{G_{*,i}},\ \mathbf{X}_{\text{MRI}_{*,j}}) \longrightarrow z_{i,j}$$

Under a linear model for quantitative traits:

$$y = G_{\text{MRI}_{*,j}}\,\beta^G_{i,j} + X\beta^X + e$$

The z-score is $z_{i,j} = \dfrac{\beta^G_{i,j}}{\text{s.e.}(\beta^G_{i,j})}$.

- $\beta^G_{i,j}$: effect size of SNP i on phenotype j
- X, β^X: covariates (age, sex, PCs...)
- p-value: $\chi^2 = \beta^2 \cdot N$, then inverted

Final Tabular ML Dataset (per illness)

Rows = SNPs, features = MRI z-scores, target = illness z-score.

$$\left[\begin{array}{cccc|c} z_{1,1} & z_{1,2} & \cdots & z_{1,m} & z_{\text{ill},1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,m} & z_{\text{ill},2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{l,1} & z_{l,2} & \cdots & z_{l,m} & z_{\text{ill},l} \end{array}\right] \begin{array}{l} \leftarrow G_1 \\ \leftarrow G_2 \\ \vdots \\ \leftarrow G_l \end{array}$$

- After clumping: $l \approx 3{,}500$ SNPs, $m \approx 1{,}000$ MRI features
- SCZ has highest predictive power (largest GWAS sample size)
- Negative median z-score is expected: allele ordering in GWAS is arbitrary

P-Value Clumping & Linkage Disequilibrium

- Nearby nucleotides are correlated (LD) because recombination is less likely at short distances.
- $\mathrm{E}[\hat{\beta}_i] = \beta_i + \sum_{j \neq i} r_{ij}\beta_j$ (LD bias within a block).
- LD blocks are estimated in PLINK.
- **Clumping**: pick the lead SNP per LD block; remove all SNPs with r^2 above a threshold with the lead SNP.
- P-value threshold (e.g. $p < 10^{-3}$) controls how many SNPs survive $\Rightarrow$ dataset size vs. signal quality trade-off.
- Same SNP set is selected across illnesses $\Rightarrow$ cross-illness comparison is valid within European ancestry.

Goal & Task Formulation

ML Objective

Learn a mapping:

$$f(\mathbf{z}_i) \approx y_i, \qquad \mathbf{z}_i = [z_{i,1}, \ldots, z_{i,m}] \in \mathbb{R}^m, \quad y_i \in \mathbb{R}$$

Two formulations:

1. **Regression**: predict the continuous illness z-score directly.
   $f : \mathbb{R}^m \to \mathbb{R}$
2. **Binary classification**: threshold the z-score $\Rightarrow$ risk / no-risk label.
   Potentially more feasible given label distribution.

Both are instances of *tabular machine learning*, which remains notoriously hard for unstructured feature sets [1], [2].

Methods

Model Landscape

Baselines

- Linear Regression
- Lasso (ℓ_1)
- Ridge (ℓ_2)
- ElasticNet

Boosted Trees

- XGBoost [3]
- LightGBM [4]
- CatBoost [5]

Deep Learning

- ResidualDNN *(ours)*
- MLP + Reg. Cocktail [1]
- TabNet [6]
- TabPFNv2 [7]

Evaluation Protocol

All models: 10 random seeds, 80/20 train/test. Metrics: R^2, MSE (regression); AUC-ROC (classification).

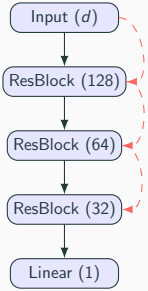
Baseline Hyperparameters

Model	Key Hyperparameter	Value
Linear Regression	–	OLS (no regularisation)
Lasso	α	0.0212 (via LassoCV)
Ridge	α	1000.0 (via RidgeCV)
ElasticNet	α, ℓ_1-ratio	0.0695, 0.1
XGBoost	`n_estimators`	200
	`max_depth`	3
	`learning_rate`	0.1
	`subsample`	0.9
	`colsample_bytree`	0.8

All models: 80/20 train/test split, 10 seeds (42–51).
Regularisation hyperparameters selected via cross-validated grid search.

ResidualDNN Architecture

- Hidden dims: $[128, 128, 64, 64, 32, 32]$
- Paired into 3 **residual blocks** with skip connections
- Each block: Linear $\rightarrow$ ReLU $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.4) $\times 2$ Residual connection
- Optimiser: Adam, lr $= 10^{-5}$
- Gradient clipping at 1.0 to prevent exploding gradients
- Epochs: 300, batch size: 32
- Learning rate scheduler: Cosine Annealing with warm restarts (T_0=15, T_mult=2)



Outlook: The *Regularisation Cocktail* [1] adds weight decay, learning-rate warmup, mixup, cutout, and label smoothing on top of a plain MLP—further avenues for improvement.

Results

Results – Schizophrenia (SCZ)

Model	Mean R^2	Mean MSE
Residual DNN	**0.409**	**13.10**
ElasticNet	0.393	13.47
Ridge	0.385	13.64
Lasso	0.382	13.70
XGBoost	0.349	14.45
Linear Regression	0.258	16.47

Table 1: Aggregated over 10 seeds (42–51), 80/20 split.

- Regularised linear models outperform plain OLS substantially ($+0.12$ R^2).
- XGBoost under-performs linear baselines—likely due to limited sample size ($\sim$3.5k).
- ResidualDNN achieves the best R^2 so far, suggesting non-linear signal is present.

Next Steps

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1. **Regularisation Cocktail MLP**
   Systematically search weight decay, mixup, cutout, learning-rate schedules, and label smoothing [1].
2. **Pre-trained TabTransformer (TabPFN-style)**
   Pre-train a transformer on MRI–GWAS data across multiple illnesses, then fine-tune on target illness. Inspired by [7].
3. **Graph Neural Network**
   Treat brain regions as nodes with MRI features; edge definition TBD (spatial adjacency, functional connectivity, or learned).
4. **Classification formulation**
   Threshold z-score into binary risk labels; evaluate with AUC-ROC alongside R^2.

References

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